

# What is HCI?

## HCI Fundamentals

Luigi De Russis, Alberto Monge Roffarello

Academic Year 2026/2027

# Hall of Fame or Shame?



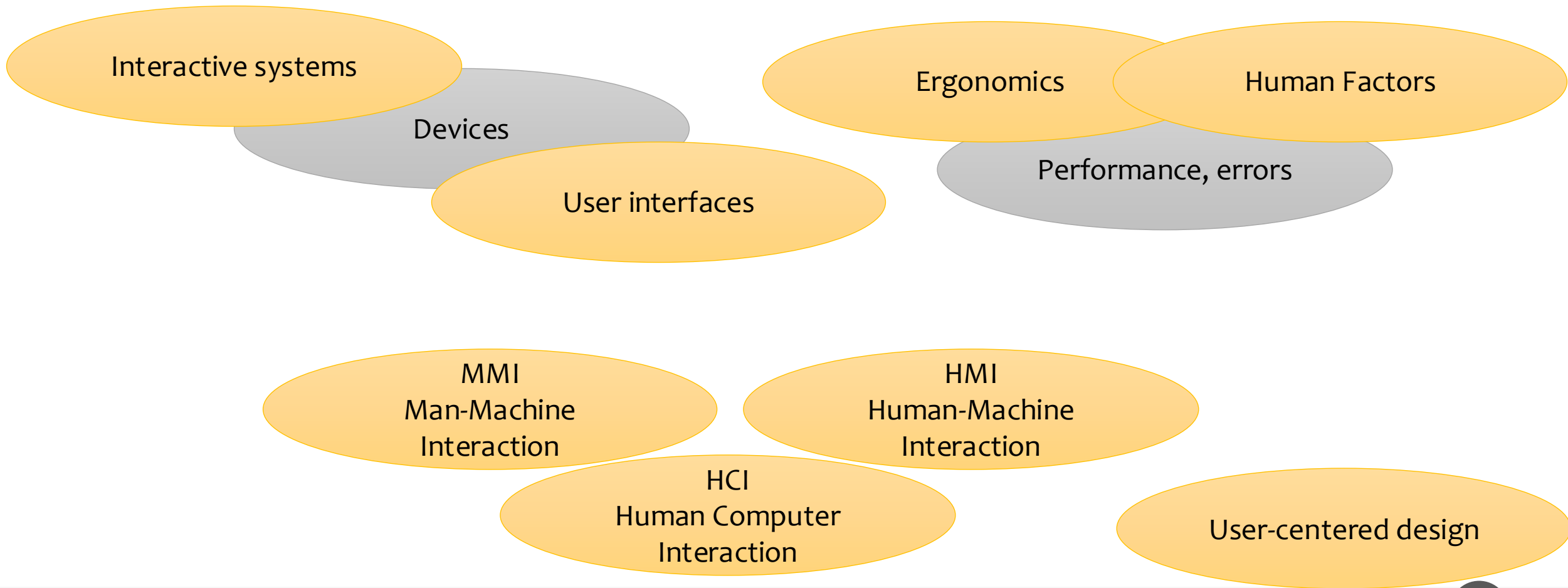
# Hall of Fame or Shame?



# Goals

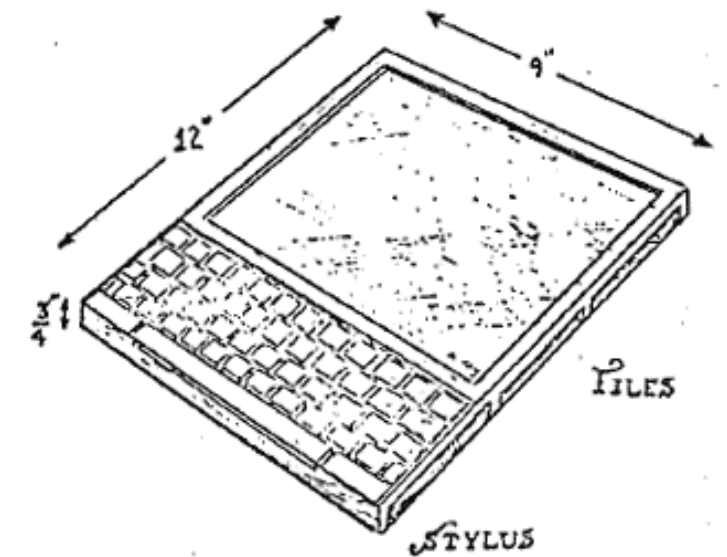
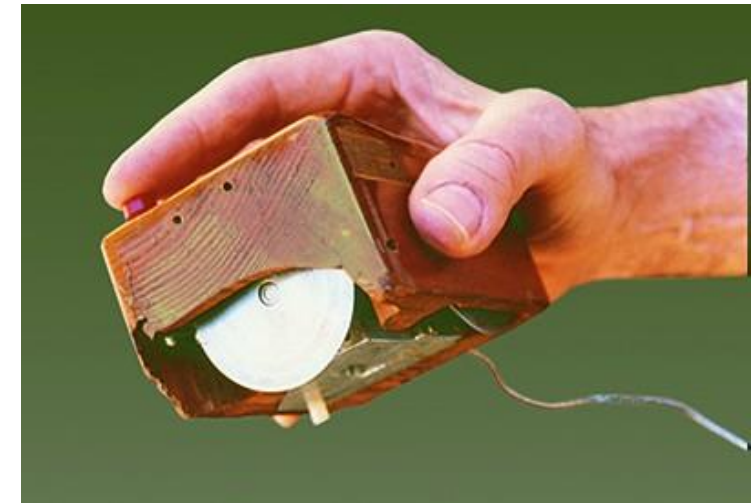
- What is HCI?
- What is usability?
- What are the Interaction Design processes, and how does they relate with Software Engineering processes?
- What is meant by User Centered Design?

# Interconnected Concepts, and Evolution



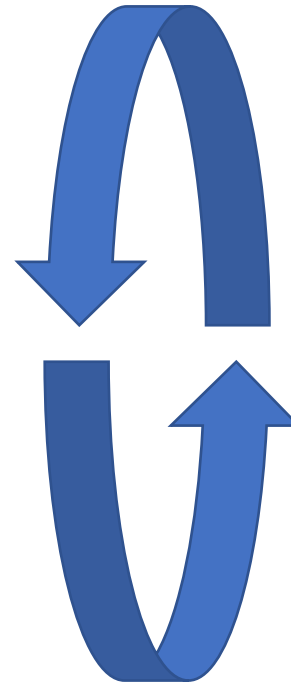
# Human-Computer Interaction... In Brief

- A multi-disciplinary field
- Concerned with the **design, evaluation, and implementation** of interactive computing systems for human use
  - and with the study of **major phenomena** surrounding them
- Involves two *entities* (the human and the computer) that determine each other behavior over time
  - framed in terms of humans' **goals** and related **tasks/pursuits**



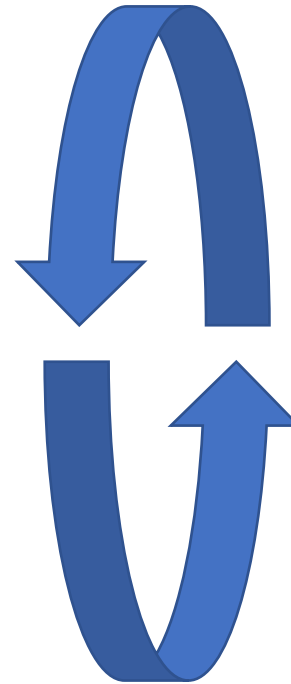
# HCI Is Multidisciplinary

- Psychology and cognitive science
  - User perceptual, cognitive and problem-solving skills
- Ergonomics
  - User's physical capabilities
- Sociology
  - Understanding the wider context of the interaction
- **Computer Science and Computer Engineering**
  - **Building the necessary artifacts (HW, SW)**
- Business
  - Satisfying market needs
- Graphic design
  - Produce an effective interface presentation
- Technical writing
  - Documentation, manuals, on-screen content
- ...



# HCI Is Multidisciplinary

- Psychology and cognitive science
  - User perceptual, cognitive and problem-solving skills
- Ergonomics
  - User's physical capabilities
- Sociology
  - Understanding the wider context of the interaction
- **Computer Science and Computer Engineering**
  - **Building the necessary artifacts (HW, SW)**
- Business
  - Satisfying market needs
- Graphic design
  - Produce an effective interface presentation
- Technical writing
  - Documentation, manuals, on-screen content
- ...



To help us in applying expertise from many different fields:

- Design methods and processes
- Models
- Heuristics
- Best practices
- Conventions
- Experiments and user studies

# The Goal of HCI

## Ingredients

- The **User(s)**
- The **Computer(s)**
- The **Task(s)** to be accomplished

## Goal

- The system must support the user's **task**, with a focus on its **usability**
  - Useful
  - Usable
  - Used

# The Ingredients

## The human

- Sensory systems
  - Visual
  - Auditory
  - Haptic
  - Spatial
- Acting systems
  - Hands
  - Voice
  - Head, Body, ...
- Cognitive processes
  - Perception
  - Memory

## The computer

- Input peripherals
  - Keyboard, mouse
  - Trackpad, trackball
  - Touch surfaces or screens
  - Microphone
  - Sensors
  - Card readers
  - ...
- Output peripherals
  - Screen
  - Audio (voice, sounds)
  - Haptics
  - VR/AR headsets
  - ...

# Models of Interaction

A general framework to understand how User and System interact

# What Is “Interaction” (in HCI)?

- Interaction is...

| Concept          | View of interaction                                              | Key phenomena and constructs                                                          | Good interaction                                                      | Example support for evaluation and design                          |
|------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------|
| Dialogue         | a cyclic process of communication acts and their interpretations | mappings between UI and intentions; feedback from the UI; turn taking                 | understandable; simple, natural; direct                               | methods/concepts for guessability, feedback, mapping; walkthroughs |
| Transmission     | a sender sending a message over a noisy channel                  | messages (bits); sender and receiver; noisy channels                                  | maximum throughput of information                                     | metrics and models of user performance                             |
| Tool use         | a human that uses tools to manipulate and act in the world       | mediation by tools; directness of acting in the world; activity as a unit of analysis | useful and transparent tools; amplification of human capabilities     | compatibility in instrumental interaction; break down analysis     |
| Optimal behavior | adapting behavior to goals, task, UI, and capabilities           | rationality; constraints; preferences; utility; strategies                            | improves or reaches maximum or satisfactory utility                   | models of choice, foraging, and adaptation                         |
| Embodiment       | acting and being in situations of a material and social world    | intentionality; context; coupling                                                     | provides resources for and supports fluent participation in the world | studies in the wild; thick description                             |
| Experience       | an ongoing stream of expectations, feelings, memories            | non-utilitarian quality; expectations; emotion                                        | satisfies psychological needs; motivating                             | metrics of user experience; experience design methods              |
| Control          | interactive minimization of error against some reference         | feedforward; feedback; reference; system; dynamics                                    | rapid and stable convergence to target state                          | executable simulations of interactive control tasks                |

**Taken from:** Kasper Hornbæk & Antti Oulasvirta, What Is Interaction? In: *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*

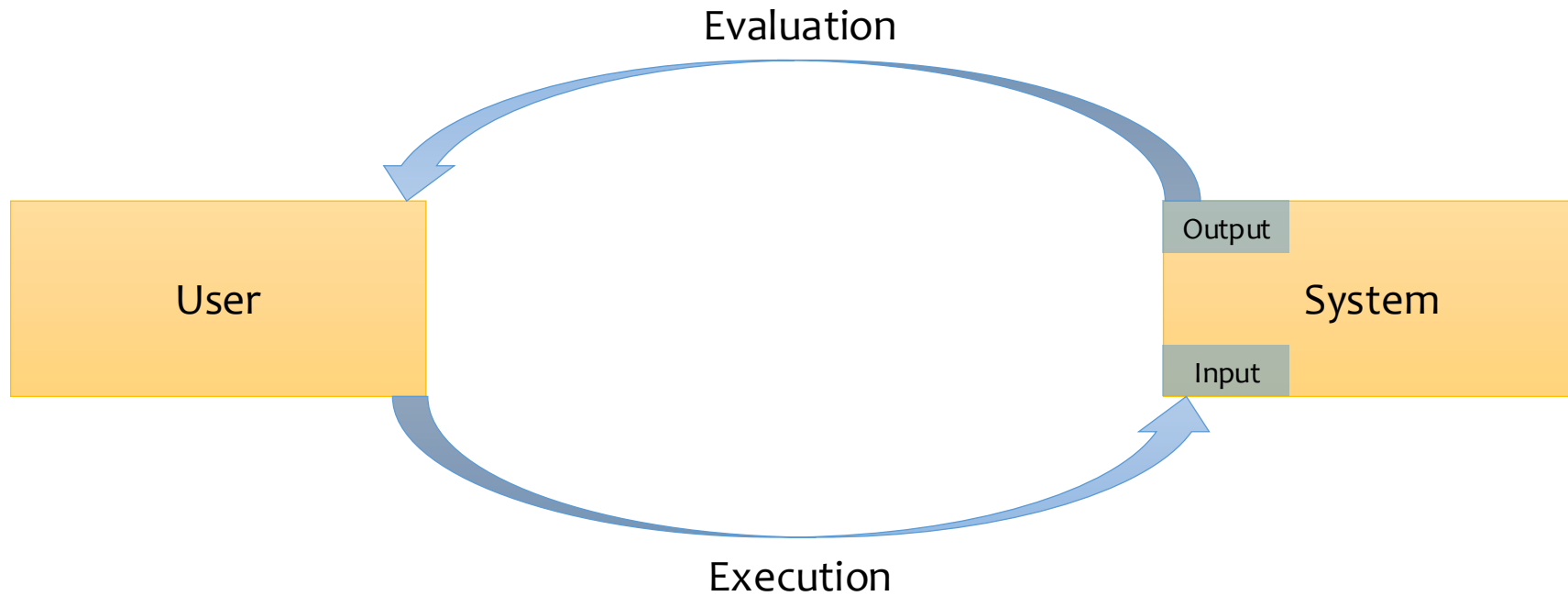
# What Is Interaction (in HCI)?

- Interaction...
  - **is not** the idea promoted and repeated in folk notions that a computer and a human are engaged
  - it concerns two entities – humans and computers – that determine each other's behavior over time
    - Their mutual determination can be of many types, including statistical, mechanical, and structural
- **Users**, with their **goals** and **pursuits**, are the ultimate metric of interaction

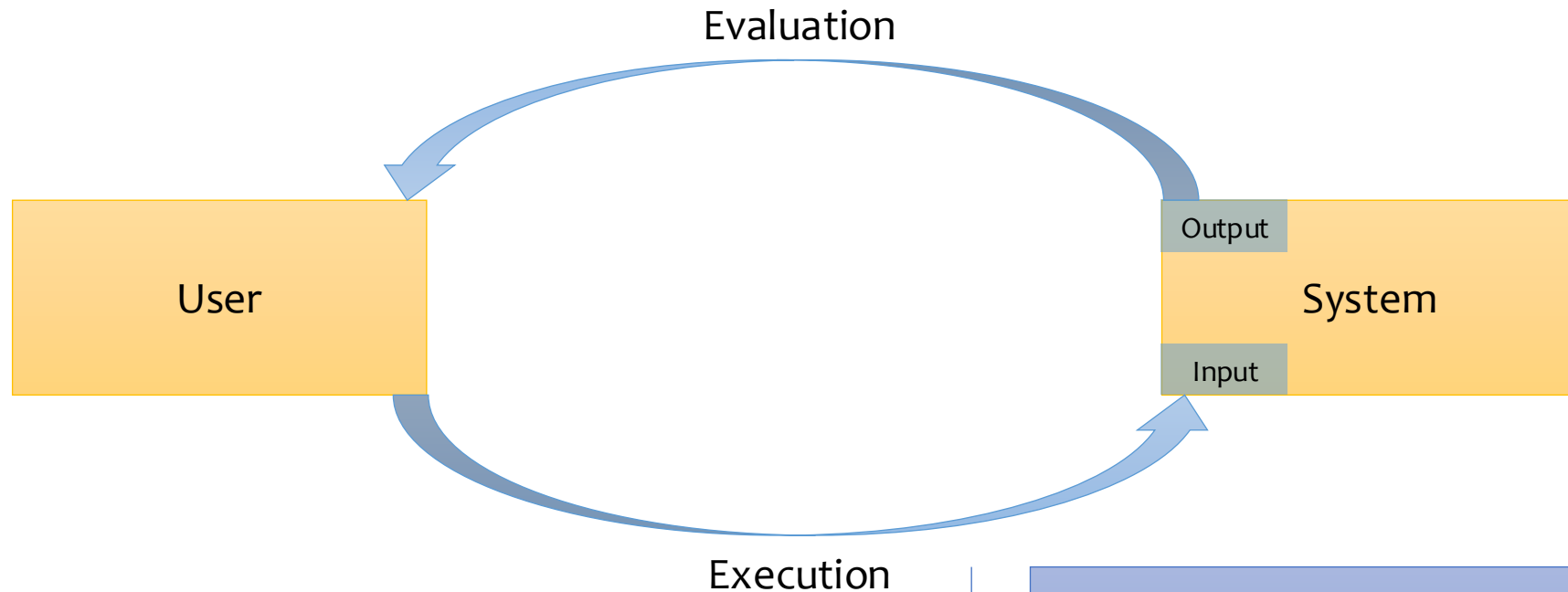
# Assumptions

- The **user** wants to accomplish some **goals**, in a specific application **domain**
  - Each domain has a specific jargon, set of possible processes and goals, artifacts and building blocks, ...
- **Tasks** are operations to manipulate the concepts of a domain
  - The goal is attained by performing one or more tasks
- Interaction studies the relation between User and System
  - The system possesses a **state** and “speaks” a **core language**
  - The user possesses a **state**, that includes an **understanding** of the system’s state, some **intention** to perform a task, and “speaks” the **task language**

# Norman's Model of Interaction

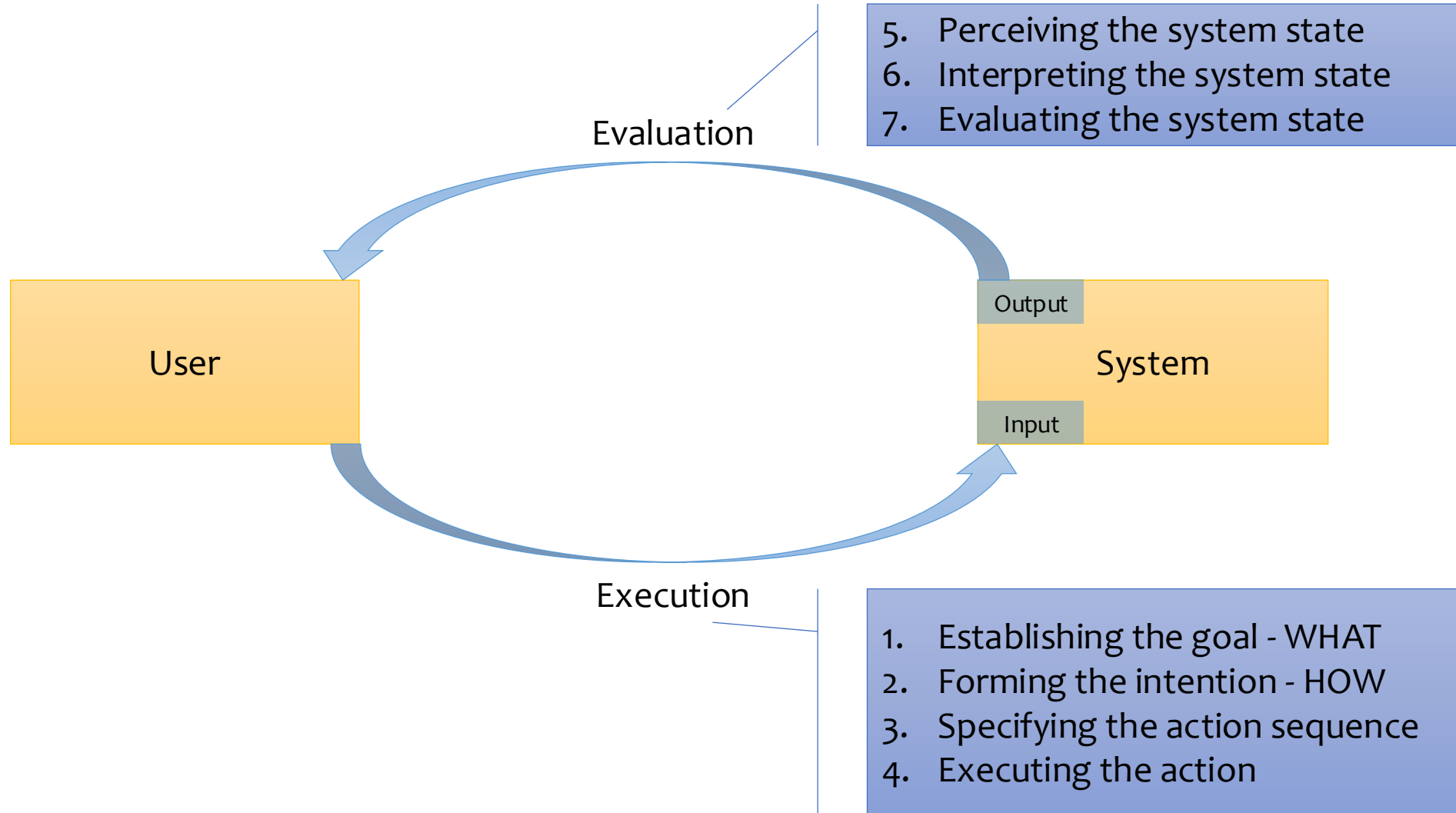


# Norman's Model of Interaction

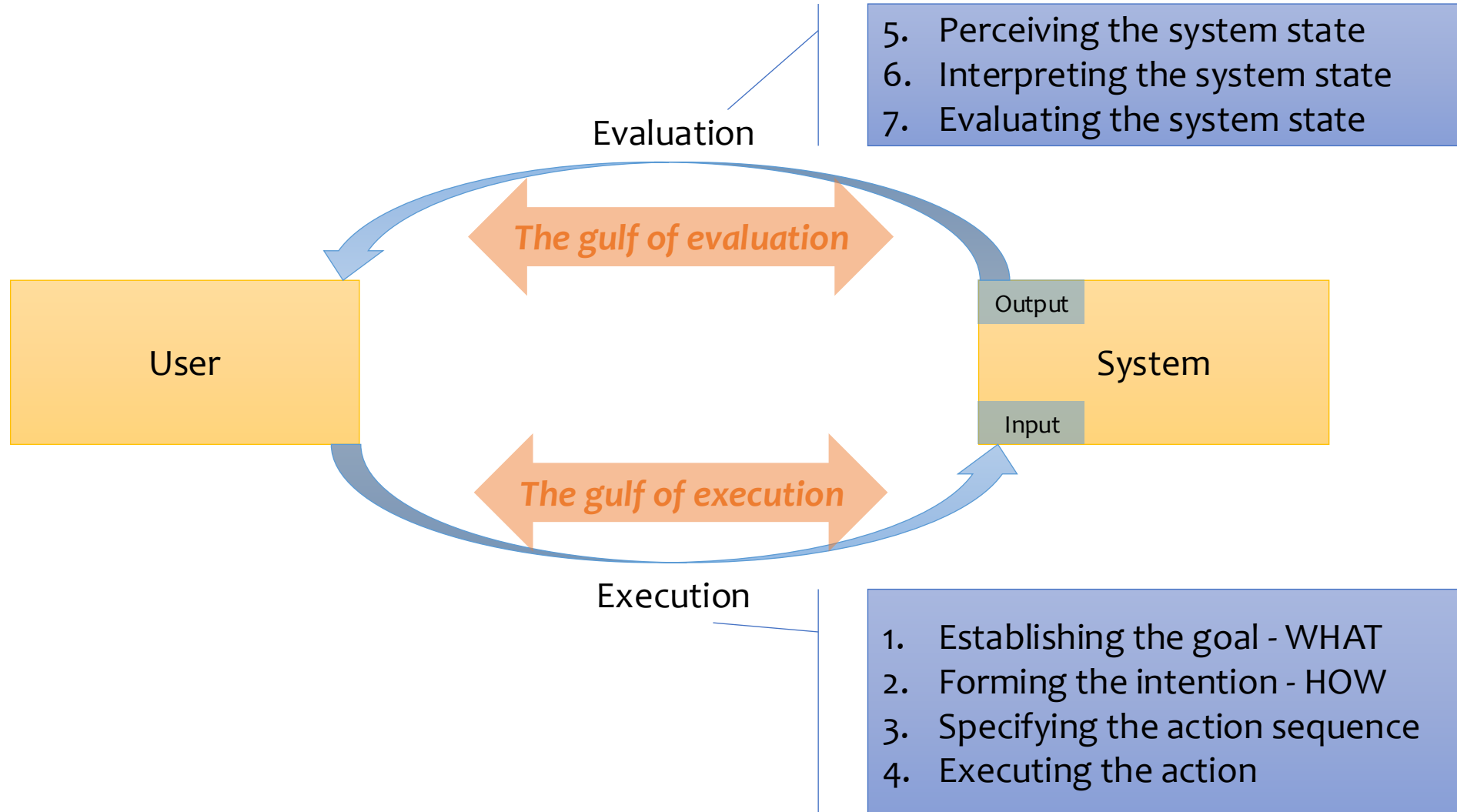


1. Establishing the goal - WHAT
2. Forming the intention - HOW
3. Specifying the action sequence
4. Executing the action

# Norman's Model of Interaction

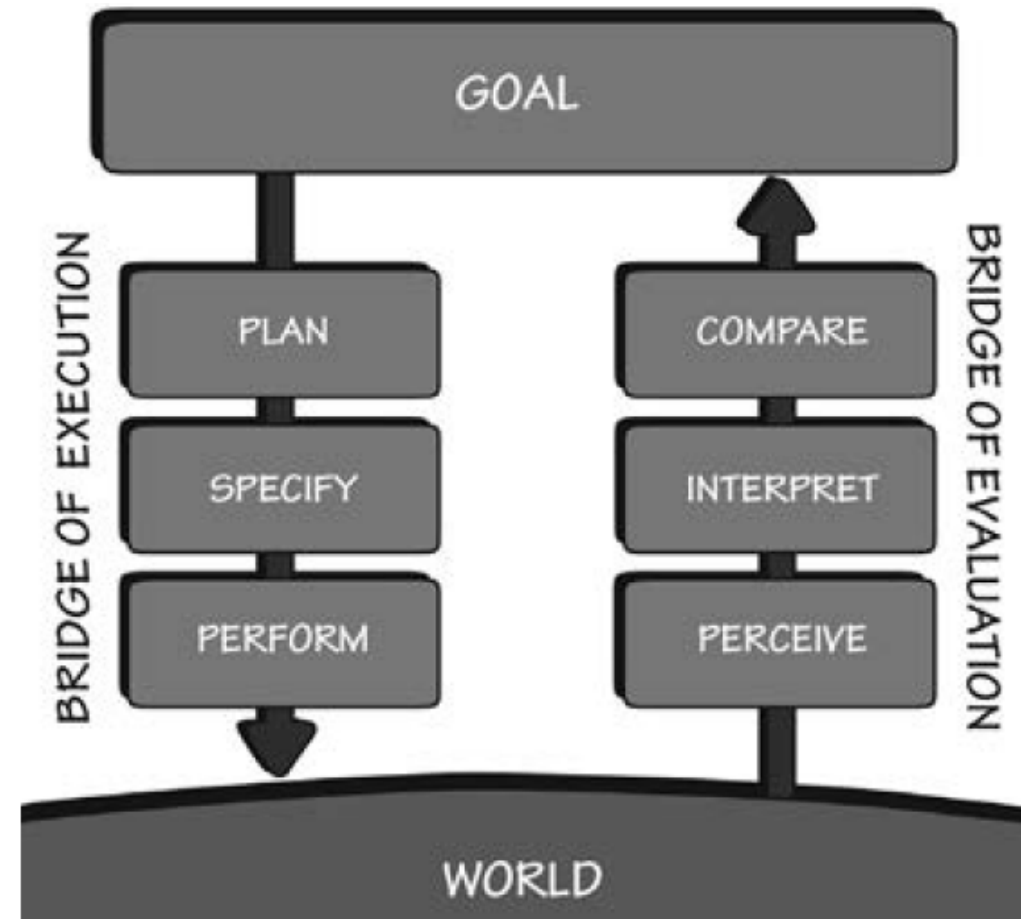
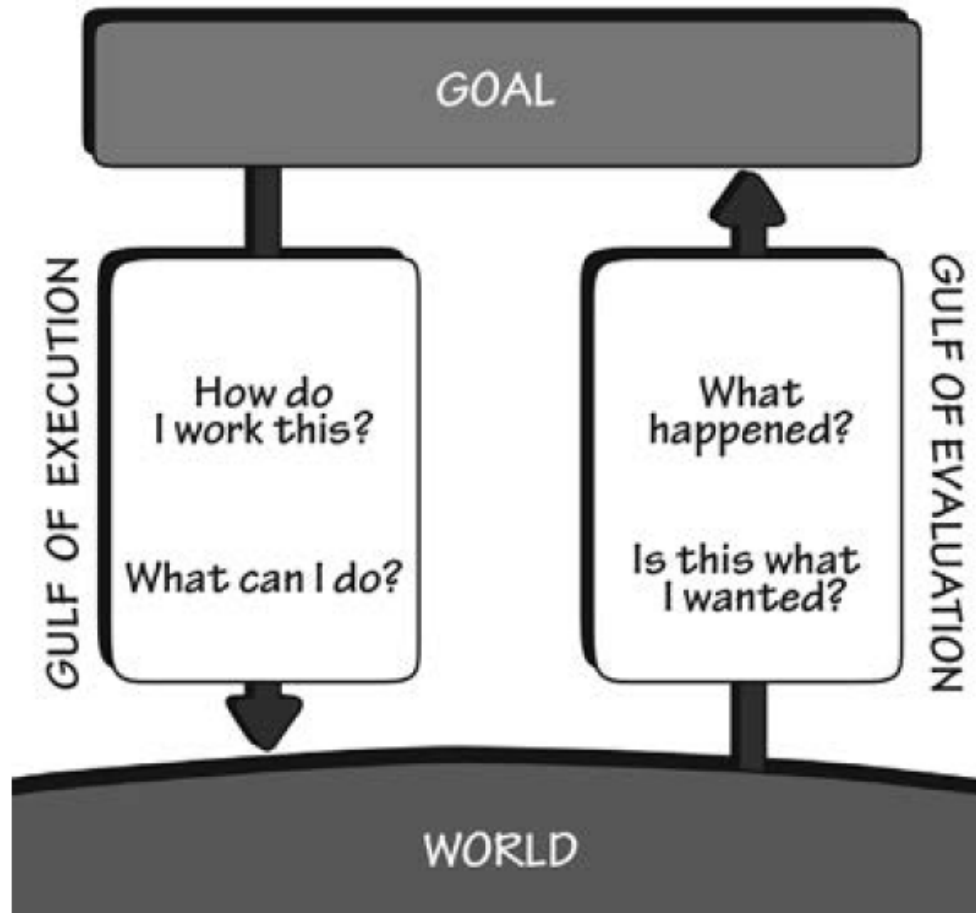


# Norman's Model of Interaction

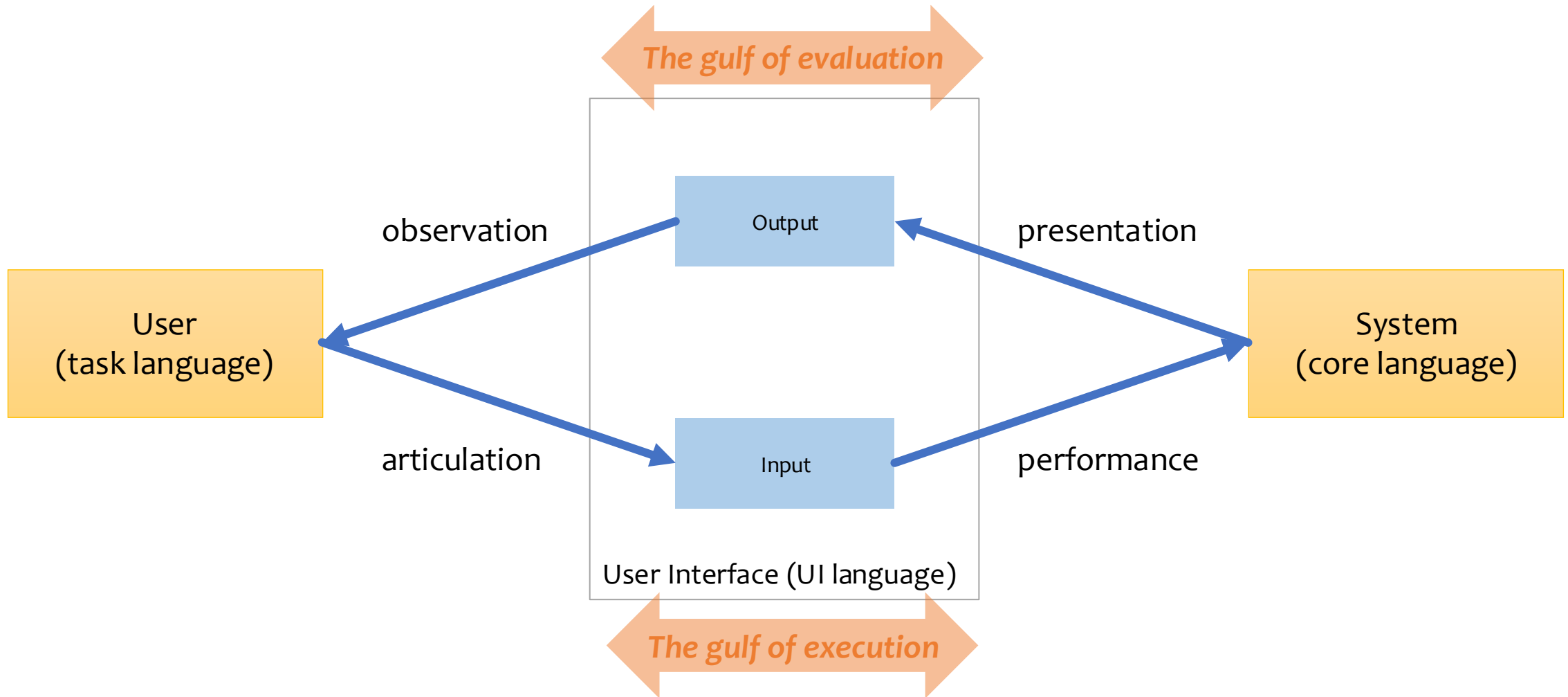


# Norman's Diagrams

1. **Goal** (form the goal)
2. **Plan** (the action)
3. **Specify** (an action sequence)
4. **Perform** (the action sequence)
5. **Perceive** (the state of the world)
6. **Interpret** (the perception)
7. **Compare** (the outcome with the goal)



# Abowd and Beale Model, with Explicit UI



# Human Errors\*

## in the gulf of execution

### Slip

- You have formulated the right action, but fail to execute that action correctly
  - E.g., click the wrong icon, or double-click too slow, ...
- May be corrected by a better interface (spacing, layout, highlights, ...)

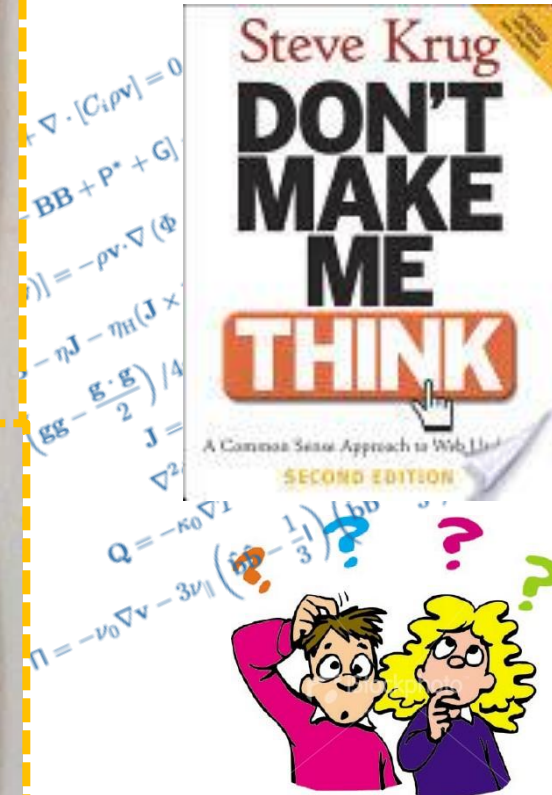
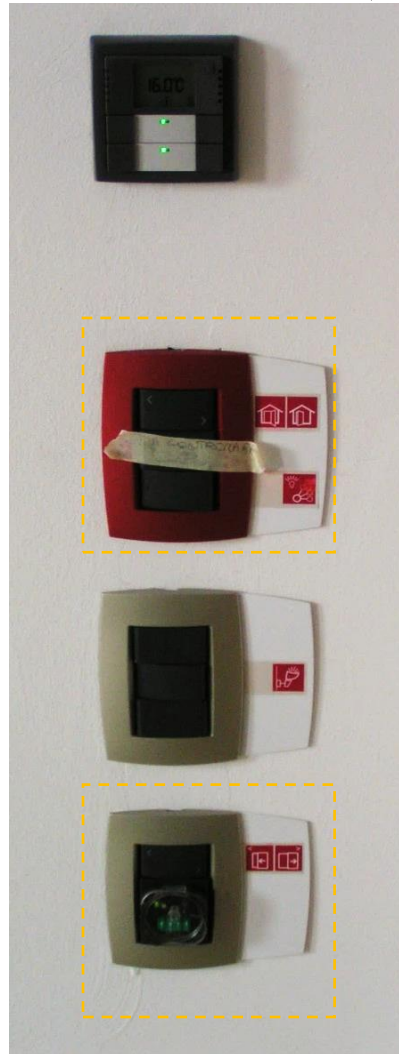
### Mistake

- You don't know the system well and you may not formulate the right goal
  - E.g., click  for Zoom, but it means Search
- The user's mental model of the system's state is not correct
- Requires more radical redesign, or additional training

## \* About Human Errors

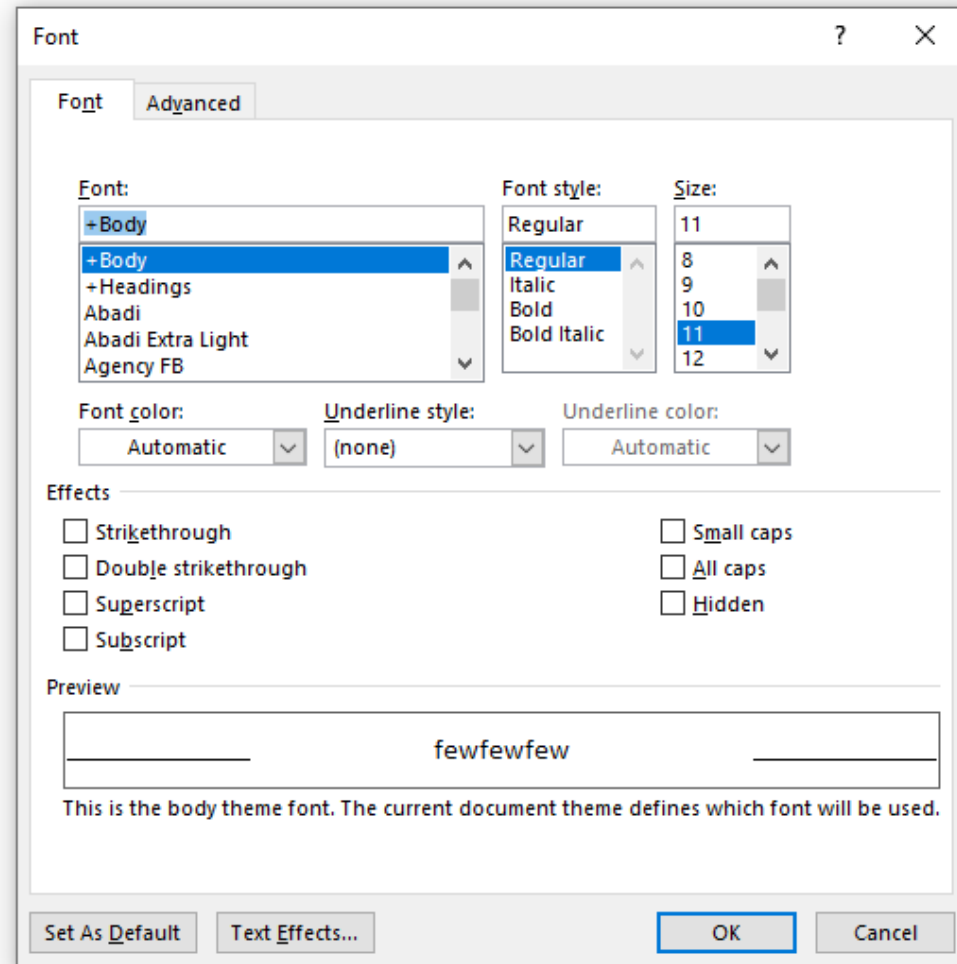
- Human errors should **never** be considered as faults of the user
- Rather, «they are usually a result of bad design» (Norman)
- Humans tend to be imprecise, distracted, not-omniscient
  - System design should anticipate this human behavior
  - Minimize the chance of inappropriate actions (evaluation)
  - Maximize the possibility of discovering and repairing an inappropriate action (execution)
  - Enable users to understand the state of the system and build an appropriate model

# Example (articulation): find the right switch

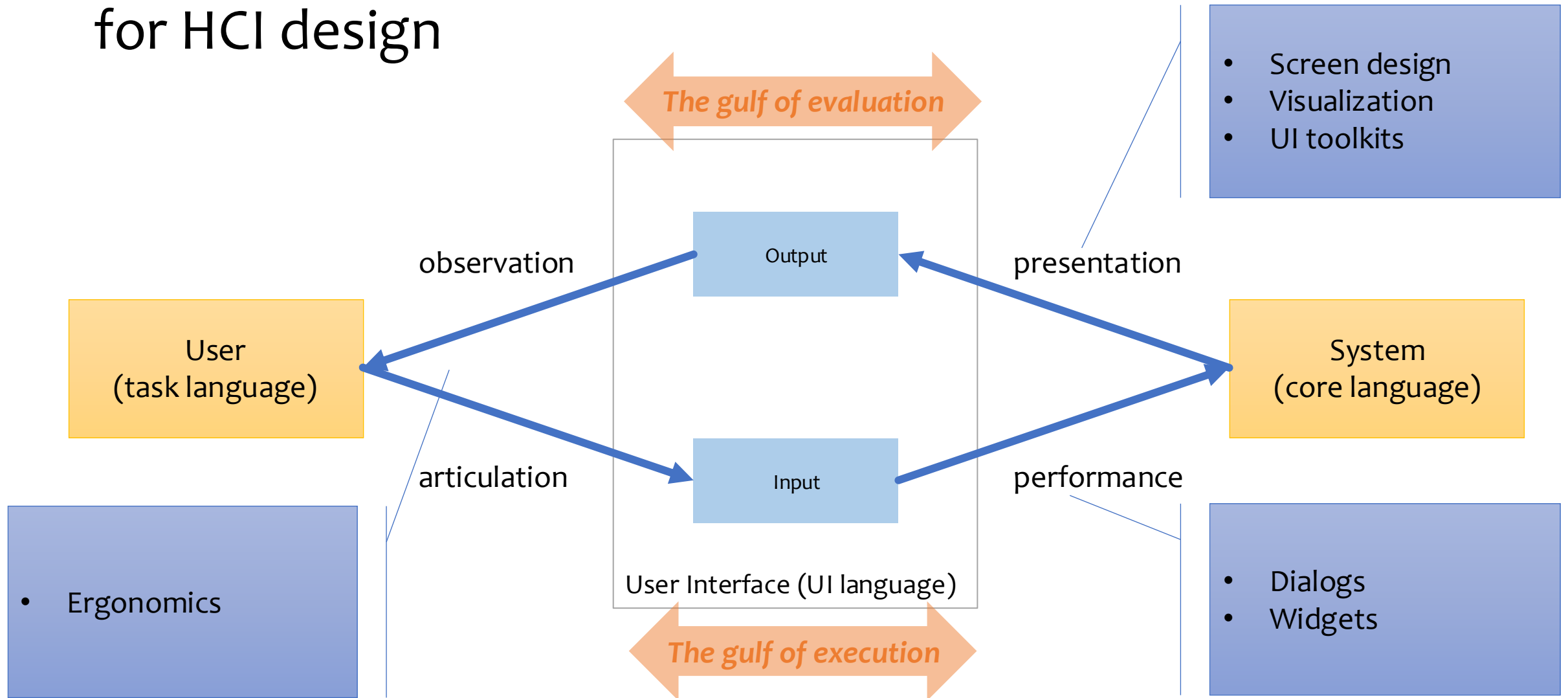


# Example (presentation):

## Which are the allowed combinations?



# Tools, Techniques and Environments for HCI design



# Frameworks: Major UI Styles

- Command line interface
- Menus
- Natural language
- Question/answer and query dialog
- Form-fills and spreadsheets
- Windows, Icons, Menus and Pointers (WIMP)
- Mobile
- Point and click
- Three-dimensional interfaces

# Design Processes and Frameworks

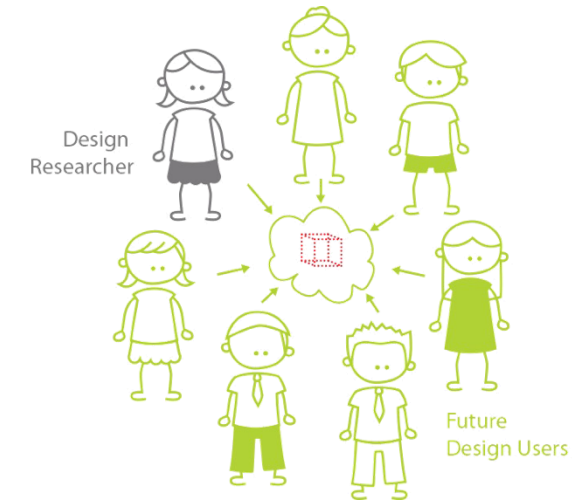
Approaches for shaping the design process

# User-Centered Design (UCD)

- Avoid the risk of software project failure
  - Estimated 50% are affected by bad developer<->user/client communication
- UCD takes the needs, wants, and limitations of the **actual end users** into account **during each phase** of the design process
  - User-centered design issues are discovered during the early stages
- Benefits: systems easier to learn, with faster performance, with less human errors, encourage users to discover advanced features, and avoids “building the wrong system”
- Issues: how to find users? How many? How motivated? How to speak their language? How to extract user needs, business needs, organizational implications?

# Participatory Design

- One step further than UCD, users are directly involved in the collaborative design of the things and applications they use
- Engage a group of users
  - Discussions
  - Creating scenarios, sketches, dramatizations
  - Creating and testing lo-fi prototypes
  - Continuous meetings, flexible management
  - Highly reliant on the skills of the group moderators/leaders (keep involved, filter ideas, reward participation, work around resistances, ...)
  - More effective with more mature and prepared user populations (less with kids, elderly, disabled, ...)



# Agile Interaction Design

- Borrows ideas from Agile development in software engineering
- Key: evolutionary development
  - System is built incrementally in rapid release cycles
  - Rapid prototyping techniques (for hardware, software, and physical objects)
- Focus on low-cost many-iterations prototypes
- Requires fast usability inspection (extreme usability, XU)
- Makers' culture (only if it involves users!)

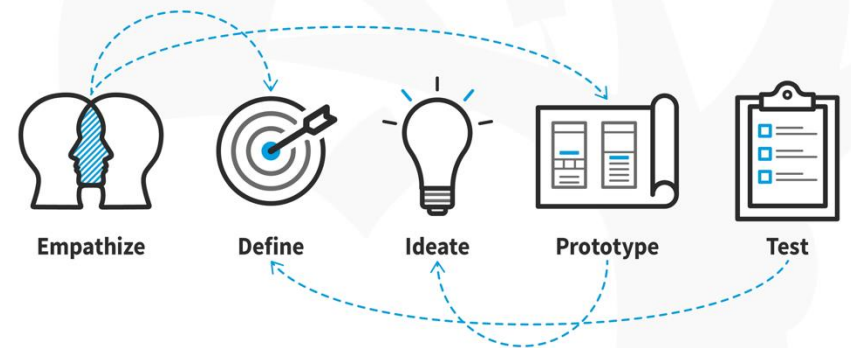
# Design Thinking

- *“A human-centered approach to innovation that draws from the designer’s toolkit to integrate the needs of people, the possibilities of technology, and the requirements for business success.”*

— [Tim Brown, IDEO’s Executive Chair](#)

- A 5-stage, non-linear and iterative, process
  1. **Empathize** – research users’ needs
  2. **Define** – state the found needs
  3. **Ideate** – challenge assumptions and ideate
  4. **Prototype** – create solutions
  5. **Test** – try the solutions out

## Design Thinking: A 5-Stage Process



Interaction Design Foundation  
[interaction-design.org](http://interaction-design.org)



# Service Design

- Describe the contemporary shift from *products* (e.g., a car of a specific brand) to *services*, e.g., the car as a tool for an elderly customer that wants to take an Uber ride to visit a friend
- Focus on the complete experience, including business resources and processes
- Build upon five *key principles*, according to “This is Service Design Thinking”:
  1. **User-centered** – focus on all users
  2. **Co-creative** – include all relevant stakeholders
  3. **Sequencing** – break a complex service into separate processes
  4. **Evidencing** – envision service, not product, experiences
  5. **Holistic** – design across networks of users and interactions

# Human-centered Design Process

A process for designing interactive systems with a focus on usability

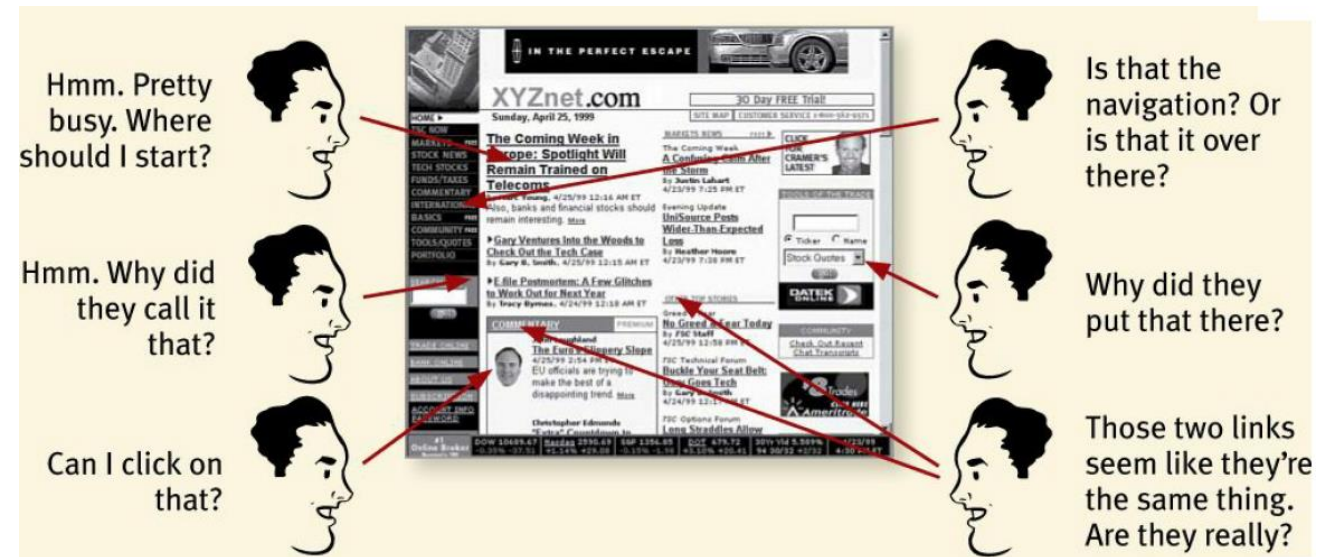
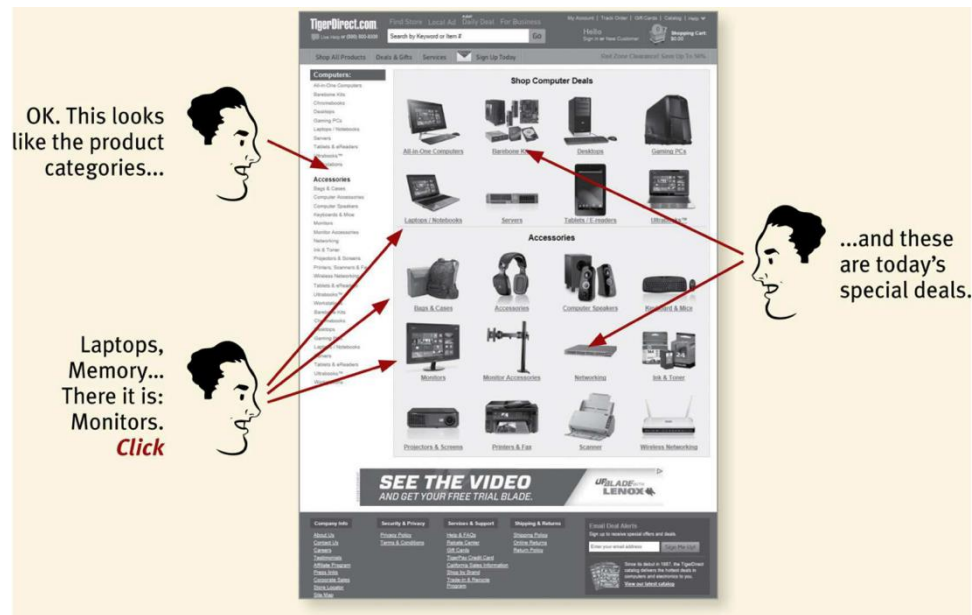
# Usability (ISO standard definition)

- **Usability:** “extent to which a system, product or service can be used by specified users to achieve specified goals with *effectiveness*, *efficiency* and *satisfaction* in a specified context of use”
  - Note 1: The “specified” users, goals and context of use refer to the particular combination of users, goals and context of use for which usability is being considered
  - Note 2: The word “usability” is also used as a qualifier to refer to the design knowledge, competencies, activities and design attributes that contribute to usability, such as usability expertise, usability engineering, usability testing, etc.

# Usability

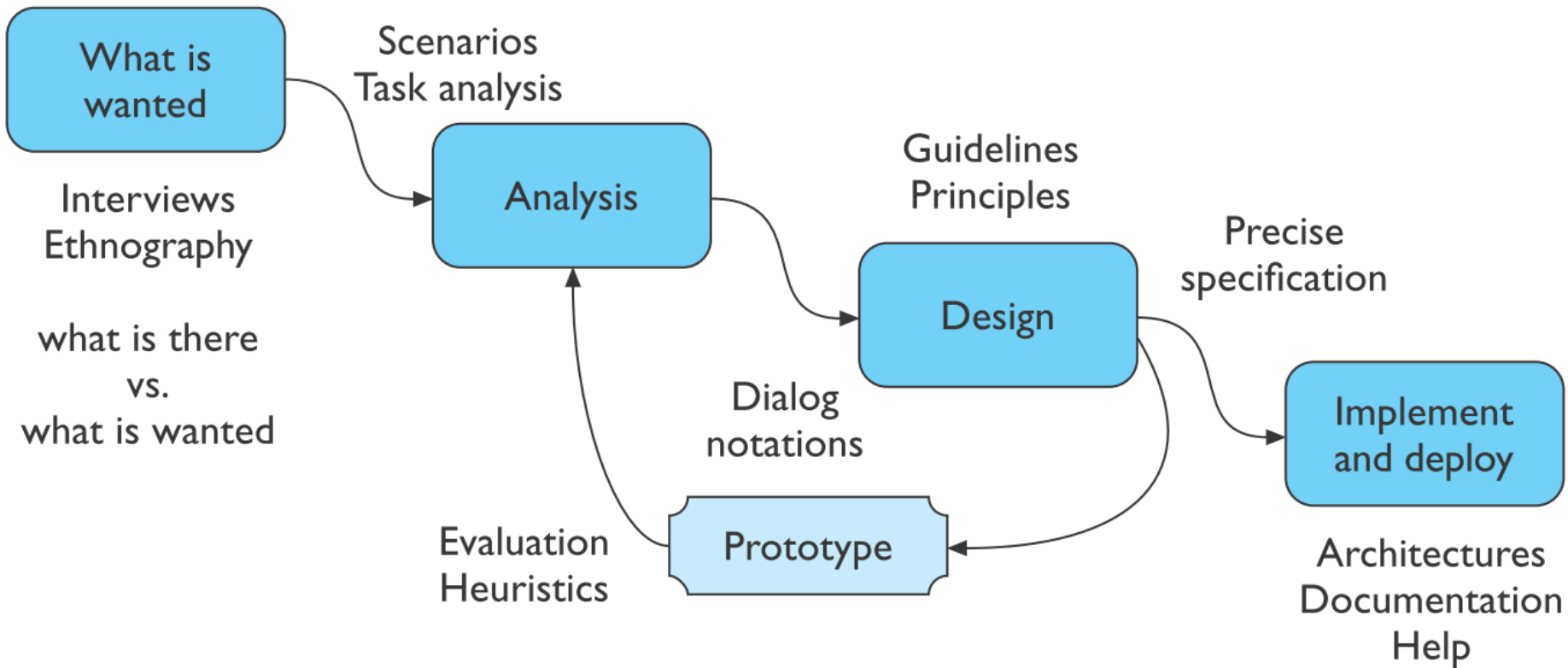
- **Usability:** how well users can use the system's functionality
- Dimensions of usability:
  - **Usefulness:** does it do something people want?
  - **Learnability:** is it easy to learn?
  - **Memorability:** once learned, is it easy to remember?
  - **Effectiveness:** does it allow reaching the goal?
  - **Efficiency:** once learned, is it fast to use?
  - **Visibility:** is the state of the system visible?
  - **Errors:** are errors few and recoverable?
  - **Satisfaction:** is it enjoyable to use?

# Usability: Don't Make Me Think



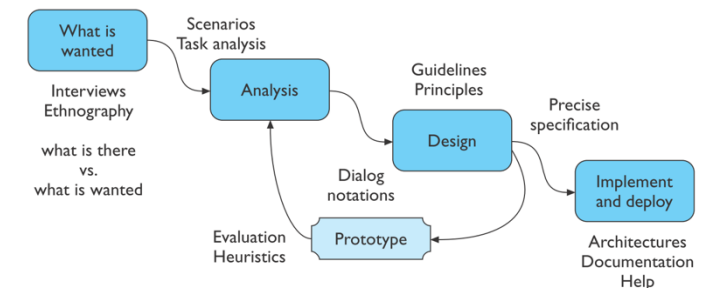
# Human-Centered Design Process

(simplified and generic)



# Human-Centered Design Process – The Main Steps (1)

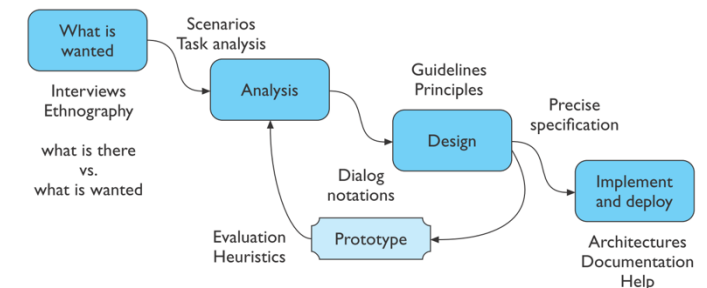
- **Needfinding** – what is needed by people
  - What exactly is needed? How are people currently accomplishing the goal?
  - User observation, interviews, ...
- **Analysis**
  - Formalize and structure the needs
  - Create interaction scenarios, stories, tasks
  - Compare current situation with expected new situation



# Human-Centered Design Process – The Main Steps (2)

## ■ Design

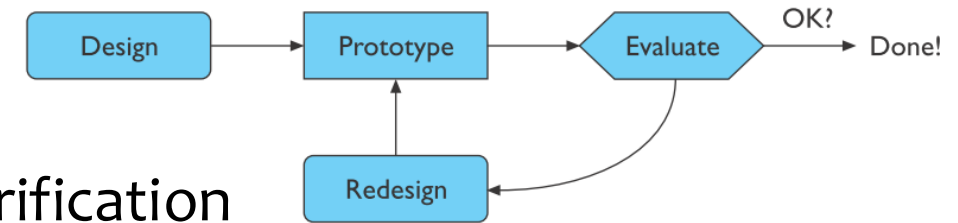
- The main choices to shape the system
- Rules, guidelines, design principles
- Considering different types of users
- Modeling and describing interaction
- Visual layout
- Consider all inputs from cognitive models, communications theories, organization issues



# Human-Centered Design Process – The Main Steps (3)

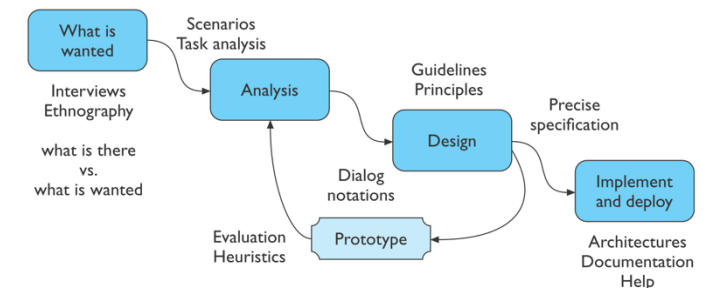
## ■ Iteration and prototyping

- Design must be supported by intermediate verification
- Evaluate the design in its partial forms:
  - Prototypes
  - Evaluation metrics
- Involving users



## ■ Implementation and deployment

- Hardware and software implementation
- Documentation

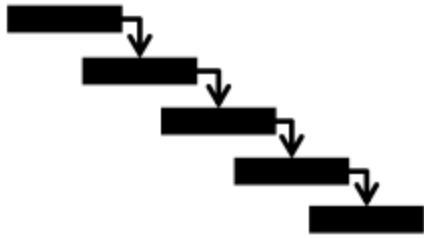


# HCI in the Software Process

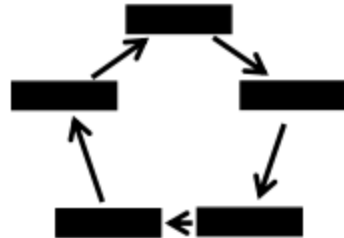
How to include Human-centered processes in Software Engineering

# Software Engineering Processes

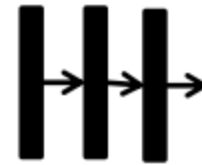
Where / how does HCI fit in?



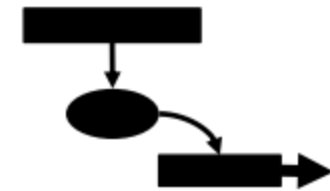
Waterfall



Iterative  
waterfall



Agile  
(scrum)

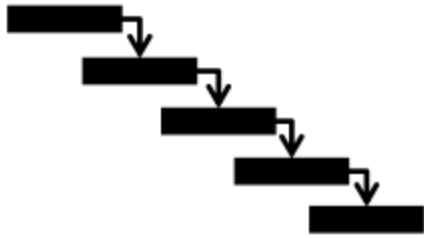


Lean

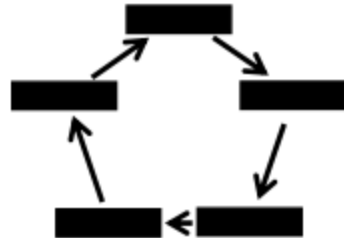
# Software Engineering Processes

Where / how does HCI fit in?

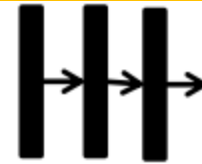
Always a step ahead!



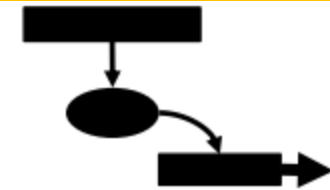
Waterfall



Iterative  
waterfall



Agile  
(scrum)



Lean

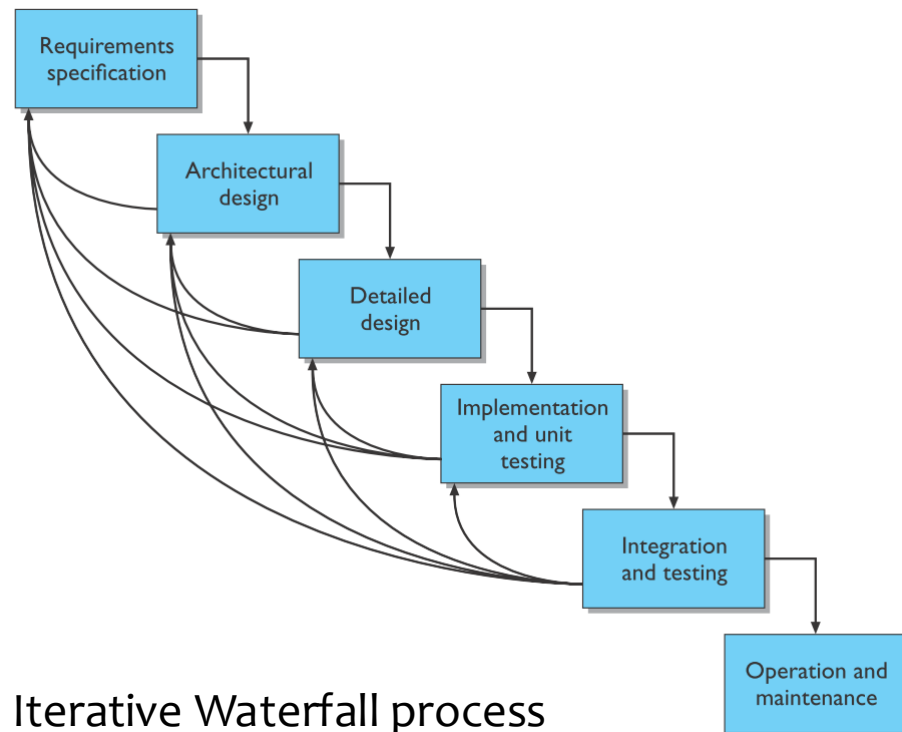
# Always a Step Ahead (I)

- Before
  - Every design step
  - Every implementation step
  - Any product iteration (or sprint)
  - ...
- You need a user-centered step
  - Evaluate usability
  - Experiment with users
  - Evaluate alternative flows
  - Evaluate alternative layouts
  - ...

# Always a Step Ahead (II)

- Usability, Safety, Performance, are part of Non-Functional Requirements
- User-centered steps are cheaper than development
  - User research about users' needs to decide what to design
  - Heuristic evaluations before testing with users
  - Evaluating prototypes instead of full-fledged products
- Anticipate critical decision points later in the project

# Example



- Each step must be
  - Preceded with user evaluation of the design choices and formalized requirements
  - Followed with user evaluation of the result
- Must produce additional artifacts to allow this kind of iteration
  - Prototypes

# References and Thanks

- Robert Miller, MIT Course “6.813/6.831: User Interface Design & Implementation”
  - Spring 2018 - <http://web.mit.edu/6.813/www/sp18/>
  - Spring 2011 - <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-831-user-interface-design-and-implementation-spring-2011/index.htm>
- Dix *et al*: Human Computer Interaction
  - Chapters 3, 5
- Norman: The Design of Everyday Things
  - Chapter 2
- Krug: Don't make me think
  - Introduction
- Shneiderman: Designing the User Interface
  - Chapter 4
- Thanks to Fulvio Corno, past teacher of the course, for his work on this slides

# License

- These slides are distributed under a Creative Commons license “**Attribution-NonCommercial-ShareAlike 4.0 International (CC BY-NC-SA 4.0)**”
- **You are free to:**
  - **Share** — copy and redistribute the material in any medium or format
  - **Adapt** — remix, transform, and build upon the material
  - The licensor cannot revoke these freedoms as long as you follow the license terms.
- **Under the following terms:**
  - **Attribution** — You must give [appropriate credit](#), provide a link to the license, and [indicate if changes were made](#). You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.
  - **NonCommercial** — You may not use the material for [commercial purposes](#).
  - **ShareAlike** — If you remix, transform, or build upon the material, you must distribute your contributions under the [same license](#) as the original.
  - **No additional restrictions** — You may not apply legal terms or [technological measures](#) that legally restrict others from doing anything the license permits.
- <https://creativecommons.org/licenses/by-nc-sa/4.0/>

